

20260323EB_MN (NCT, ATP)

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2026年3月23日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2026年3月25日 星期三

T2 -> T3

2026年3月27日 星期五

T3 -> T4 (~14 mL for over-weekend)

2026年3月30日 星期一

T4 -> T5

2026年4月1日 星期三

T5 -> T6

2026年4月3日 星期五

T6 -> T6 (~14mL for over-weekend)

2026年4月6日 星期一

T6 -> T7

Dissociation

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 4mL 0.25% trypsin (prepare 8mL 0.25% trypsin with 220U DNaseI)
3. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
4. pipette*10 at 5min, 10min, and 15min
5. add 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
6. add 8mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.
8. centrifuge at 200xg, 4min

cell info1 ^

	A	B	C
1	cell	passage	well#
2	WT11	P6	5

Table2 ^

	A	B
1		volume (mL)
2	trypsin	4
3	HI-FBS/CTWM	8

Table3 ✓

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	3.38	99	6	20.28	A	24	250	24	1775

Well1 ^

	1	2	3	4	5	6
A	110 (12uL)	111 (18uL)	92(15uL in total)			92 5uL + 105 2uL
B						
C						
D						

2026年4月7日 星期二

Change T7 -> T7

2026年4月9日 星期四

T7 -> T8 (aspirate out the old media with suction. 110 and 111 seemed a bit stressed, so some cells detached. I think the virus titer became higher with the late aliquot.)

2026年4月11日 星期六

1/2 T8 -> 1/2 T8 (with Theo's help)

2026年4月13日 星期一

1/2 T8 -> 1/2 T8

2026年4月15日 星期三

1/2 T8 -> 1/2 T8

Imaging NCT

Prepare 2.5mM gluc NBA with full supps

8mL: 8mL no gluc NBA with full supps + 18.2uL 1.1M glucose

take 4mL + 0.8uL SirDNA (1:5000)

drug calculation												
	A	B	C	D	E	F	G	H	I	J	K	L
1		stock (uM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	working (uM)	dilution factor from 1st dil.	total volume	volume of 1st dil.	volume of media
2	MMS	11800000	100	100	1	99	118000	500	236	1000	4.2	995.8
3		11800000	100	100	1	99	118000	1000	118	1000	8.5	991.5
4		11800000	100	100	1	99	118000	3000	39.33333333333	1000	25.4	974.6

NCT plate layout						
	1	2	3	4	5	6
A	1	1				
B	2	2				
C	3	3				
D	4	4				

Table1		
	A	B
1	condition#	condition
2	1	0mM MMS
3	2	0.5mM MMS
4	3	1mM MMS
5	4	3mM MMS

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	

time table for NCT treatment1										
	A	B	C	D	E	F	G	H	I	J
1	NCT									
2	condition#		condition	media	time	imaging	imaging order			
3	on 4/18/2026		110							
4	1	A1	0mM MMS	2.5mM gluc	10:00	11:05		1		
5	2	B1	0.5mM MMS	2.5mM gluc	10:30	11:35		2		
6	3	C1	1mM MMS	2.5mM gluc	11:00	12:01		3		
7	4	D1	3mM MMS	2.5mM gluc	11:30	12:34		4		
8			111							
9	1	A2	0mM MMS	2.5mM gluc	12:30	13:32		5		
10	2	B2	0.5mM MMS	2.5mM gluc	13:00	14:02		6		
11	3	C2	1mM MMS	2.5mM gluc	13:30	14:30		7		
12	4	D2	3mM MMS	2.5mM gluc	14:00	15:00		8		

fileID_treatment_NCT				
	A	B	C	D
1	image	reporter	treatment	time
2	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	1h
3	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	0mM MMS	1h
4	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.5mM MMS	1h
5	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0.5mM MMS	1h
6	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	1mM MMS	1h
7	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	1mM MMS	1h
8	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	3mM MMS	1h
9	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	3mM MMS	1h
10	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	1h
11	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	0mM MMS	1h
12	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.5mM MMS	1h
13	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0.5mM MMS	1h
14	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	1mM MMS	1h
15	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	1mM MMS	1h
16	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	3mM MMS	1h
17	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	3mM MMS	1h

2026年4月16日 星期四

Imaging: ATP/ADP ratio (Perceval Sensor)

10mM lactate (10mM lactate: 224.2uL of 446mM lactate in 10mL media)

1mM pyruvate

no pyruvate or lactate

Well2						
	1	2	3	4	5	6
A			F	D	C	F (2.5mM gluc)
B			A	E	D	H (0mM gluc)
C			B	A	E	G (0mM gluc)
D			C	B	F	F (2.5mM gluc)

prepare media with different conc. of 2DG							
	A	B	C	D	E	F	G
1	condition #	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	A	2000	50	40	1000	25.0	975.0
3	B	2000	20	100	1000	10.0	990.0
4	C	2000	10	200	1000	5.0	995.0
5	D	2000	6	333	1000	3.0	997.0
6	E	2000	2	1000	1000	1.0	999.0
7	F		0		1000	0.0	1000.0

prepare media with different conc. of glucose							
	A	B	C	D	E	F	G
1	condition #	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	F	1100	2.5	440	1000	2.3	997.7
3	G		0		1000		
4	F	1100	2.5	440	7000	15.9	6984.1
5	G		0				
6	F	1100	2.5	440	6000	13.6	5986.4

O.C. , 60X, zoom1.5, 600x900, 16-bit, Ti2 (new scope)					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	
2	DIC				
3	488	16	20	200	
4	Dapi-ex-Green-em	16	20	300	
5	560	16	30	400	column6 only. signal was bad

time table for ATP treatment																			
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Perceval sensor			with 1mM pyruvate						without pyruvate or lactate									
2	#		condition	media	time	imaging		#		condition	media	time	imagin g		#		condition	time	imagin g
3	F	A3	0mM 2DG	2.5mM gluc	11:00	12:00		A	C4	50mM 2DG	2.5mM gluc	13:20	14:20		F	A6	2.5mM gluc w/ 1mM pyruvate	15:03	16:00
4	A	B3	50mM 2DG	2.5mM gluc	11:15	12:15		B	D4	20mM 2DG	2.5mM gluc	13:30	14:30		G	B6	0mM gluc w/ 1mM pyruvate	15:20	16:15
5	B	C3	20mM 2DG	2.5mM gluc	11:30	12:34		C	A5	10mM 2DG	2.5mM gluc	13:45	14:45		G	C6	0mM gluc w/ 10mM lactate	15:32	16:33
6	C	D3	10mM 2DG	2.5mM gluc	11:45	12:45		D	B5	6mM 2DG	2.5mM gluc	14:00	15:00		F	D6	2.5mM gluc w/ 10mM lactate	15:45	16:45
7	D	A4	6mM 2DG	2.5mM gluc	12:03	13:00		E	C5	2mM 2DG	2.5mM gluc	14:17	15:17						
8	E	B4	2mM 2DG	2.5mM gluc	12:18	13:18		F	D5	0mM 2DG	2.5mM gluc	14:32	15:30						

~~Not totally sure if pHred works since the signal of 560 is associated to 488 signal (but some of them are not).~~

--> I think the signal is true since I checked D5 and it doesn't have 560 signal. In column 6, every cell has signal. just not bright.

There are still many puncta in the cells with 560, which 488 doesn't have.

fileid_treat_ATP1				
	A	B	C	D
1	image	treatment	treat_time	pyruvate
2	vLH92_MN_60x-1xzoom_0000	0mM 2DG	1h	yes
3	vLH92_MN_60x-1xzoom_0001	50mM 2DG	1h	yes
4	vLH92_MN_60x-1xzoom_0002	20mM 2DG	1h	yes
5	vLH92_MN_60x-1xzoom_0003	10mM 2DG	1h	yes
6	vLH92_MN_60x-1xzoom_0004	6mM 2DG	1h	yes
7	vLH92_MN_60x-1xzoom_0005	2mM 2DG	1h	yes
8	vLH92_MN_60x-1xzoom_0006	50mM 2DG	1h	no
9	vLH92_MN_60x-1xzoom_0007	20mM 2DG	1h	no
10	vLH92_MN_60x-1xzoom_0008	10mM 2DG	1h	no
11	vLH92_MN_60x-1xzoom_0009	6mM 2DG	1h	no
12	vLH92_MN_60x-1xzoom_0010	2mM 2DG	1h	no
13	vLH92_MN_60x-1xzoom_0011	0mM 2DG	1h	no
14	vLH92_MN_60x-1xzoom_0012	2.5mM gluc	1h	yes
15	vLH92_MN_60x-1xzoom_0013	0mM gluc	1h	yes
16	vLH92_MN_60x-1xzoom_0014	0mM gluc	1h	10mM lactate
17	vLH92_MN_60x-1xzoom_0015	2.5mM gluc	1h	10mM lactate

